SREENIDHI UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Deep Learning Real-Time Mini-Project | Academic Submission | AY: 2025–2026

PROJECT TITLE: Deep Learning-Based Detection of AI-Generated and Real Images Using Convolutional Neural Networks

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1. MANDATORY GITHUB REPOSITORY DETAILS

GitHub Repository URL	**https://github.com/LalithaSreya/DL-Mini-Project-AI-Image-Detection (Default Branch: main)**
Core Executables	project.ipynb (Google Colab 1-Click with GPU), app.py (Interactive Streamlit Web UI), main.py (Training & Eval pipeline), predict.py (Single-Image inference)
Project Deliverables	docs/Project_Report.pdf (Full 20+ page academic report), docs/Project_Report.docx, results/figures/ (Confusion matrix & accuracy/loss curves), results/metrics/
Quick Run Instructions	git clone https://github.com/LalithaSreya/DL-Mini-Project-AI-Image-Detection.git pip install -r requirements.txt | streamlit run app.py

2. DATASET SPECIFICATIONS, SOURCE & ATTRIBUTION

Dataset Name & Scope	CIFAKE: Real and AI-Generated Synthetic Images — 120,000 images (32x32 pixels, RGB)
Class Breakdown	REAL (60,000 CIFAR-10 authentic images) vs. FAKE (60,000 Stable Diffusion v1.4 synthetic images across 10 semantic classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck)
Partition Splits	Training Split: 100,000 images (50,000 REAL / 50,000 FAKE) | Test Split: 20,000 images (10,000 REAL / 10,000 FAKE)
Kaggle Dataset URL	**https://www.kaggle.com/datasets/birdy654/cifake-real-and-ai-generated-synthetic-images**
Citation & Open Data	Bird, J. J., & Lotfi, A. (2023). 'CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images.' IEEE Access, 12, 15600-15612. Aligned with Govt. of India AIKosh platform (https://aikosh.indiaai.gov.in/) & fast.ai practical deep learning standards.

3. DEEP LEARNING MODEL & EVALUATION RESULTS

Test Accuracy	Macro F1-Score	Test Binary Loss	Deep Architecture
93.93%	**93.93%**	**0.1639**	**MobileNetV2 (GAP + Drop 0.3)**